GGG-P-441a
April 9, 1965

SUPERSEDING
Fed. Spec. GGG-P-441
August 3, 1956

FEDERAL SPECIFICATION

PLATES, ANGLE, ADJUSTABLE AND SOLID

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers angle plates or knees for inspection, layout and machine operations.

1.1.1 *Federal specification coverage.* This specification covers only the types and classes of plates generally purchased by the Federal Government and is not intended to include all of the types and classes which are commercially available.

1.2 Classification.

1.2.1 *Types and classes.* Angle plates or knees shall be of the following types and classes, as specified (see 6.2):

Type I—Right-angle plates or knees.
- Class 1—Plain.
- Class 2—Slotted.
- Class 3—Measuring, plain.

Type II—Inside right-angle plates.

Type III—Universal right-angle plates.

Type IV—Machinist's adjustable-angle plates.

Type V—Toolmaker's adjustable knee.

Type VI—Toolmaker's nonadjustable knee.

1.2.2 *Grades.* Angle plates or knees covered by this specification shall be of the following grades as specified (see 3.6 and 6.2):

Grade A—Fine scraped.

Grade B—Ground.

Grade C—Fine machined.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

PPP-B-640—Boxes, Fiberboard, Corrugated, Triple Wall.

PPP-F-320—Fiberboard; Sheet, Stock and Cut Shapes.

PPP-T-76—Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 151—Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index.

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The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-C-104—Crates, Wood, Lumber and Plywood Sheathed, Nailed and Bolted.

MIL-P-116—Preservation, Method of.

MIL-P-3420—Packaging Materials, Volatile Corrosion Inhibitor, Treated, Opaque.

Military Standards:

MIL-STD-10—Surface Roughness, Waviness and Lay.

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal shall apply.

National Bureau of Standards:

Handbood H-28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are descriptive and not restrictive and are not intended to preclude the purchase of angle plates or knees otherwise conforming to this specification.

3.2 Construction. Construction shall be free from any characteristics or defects that may render the angle plate or knee unsuitable or inefficient for the purpose intended.

3.3 Castings. Angle plates or knees covered by this specification shall be made from high grade, fine-grained, controlled process, iron or iron alloy castings. Castings shall be of uniform quality, free from blowholes, porosity, hard spots, shrinkage and other injurious defects. The castings shall be thoroughy cleaned of all other scale, molding sand and other extraneous materials. Flaws in castings shall not be plugged, peened, welded, or otherwise repaired without giving full particulars to the extent of the repair to the procuring agency. Acceptance of repaired castings shall be at the option of the procuring agency.

3.3.1 *Painting.* All surfaces not machined shall be given one coat of primer and two coats of lacquer or enamel. Unless otherwise specified (see 6.2), color shall be manufacturer's standard.

3.4 Seasoning. After being cast and rough machined, each angle plate or knee shall be stress relieved to insure dimensional stability.

3.5 Hardness. The finished angle plate or knee shall have a Brinell hardness of not less than 180 when using a 10 millimeter (mm.) hardened-steel ball with a 3,000 kilogram load.

3.6 Grades. Angle plates of knees covered by this specification shall be of the grades specified in 3.6.1 through 3.6.3 (see 6.2).

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3.6.1 *Grade A.* Grade A angle plates or knees shall have working surfaces that are fine precision scraped with relief spots to prevent sticking of instruments and gage blocks. The spotting cavities of the working surface shall be from 0.0002 inch to 0.0005 inch in depth with a concentration of 15 to 18 spotting cavities per square inch and a 20 to 40 percent bearing. The bearing area shall not deviate from the mean plane by more than that specified in table I. No adjacent square areas shall vary with each other by more than 0.0001 inch per square foot. The adjacent working surfaces of angle plates or knees shall not vary from square by more than 0.0001 inch in every 8 inches.

3.6.2 *Grade B.* Grade B angle plates or knees shall have working surfaces that shall be precision ground with a r.m.s. value of not more than 16 microinches for angle plates or knees with less than 64 square inches per working surface nor more than 32 microinches for the largest size and shall be in accordance with MIL-STD-10. The bearing area of each working surface shall not deviate from the mean plane by more than that specified in table I. No adjacent square areas shall vary with each other by more than 0.0001 inch per square foot. The adjacent working surfaces of angle irons and knees shall not vary from square by more than 0.0001 inch in every 8 inches.

3.6.3 *Grade C.* The working surfaces of grade C angle plates or knees shall be smooth machined and shall not exceed a surface roughness of 32 r.m.s. in accordance with MIL-STD-10. The bearing area of each working surface shall not deviate from the mean plane by more than that specified in table I. No adjacent square foot areas shall vary with each other by more than 0.0003 inch per square foot. The adjacent working surfaces of angle irons and knees shall not vary from square by more than 0.0003 inch in every 8 inches.

TABLE I. *Permissible deviations*

Maximum dimension of working surface		Maximum permissible deviation of bearing areas from mean plane	
Above	To and including	Grades A and B	Grade C
Inches	*Inches*	*Inch*	*Inch*
2-1/2	12	0.0001	0.0002
12	24	.0001	.0002
24	36	.0002	.0004
36	48	.0002	.0006
48	60	.0002	.0008

3.7 **Sizes.** Angle plates or knees covered by this specification shall be of the sizes specified hereinafter in tables II through VIII (see 6.2).

3.8 **Ribbing.** Ribbing, of sufficient thickness, shall be incorporated into the design of angle plates or knees to assure maximum flatness, squareness, and stability of all working surfaces.

3.9 **Threads.** Threads incorporated in the construction of types IV and V shall conform to H28.

3.10 **Type I, right-angle plates or knees.**

3.10.1 *Class 1, plain.* Class 1 angle plates or knees shall have two members with the outside working surfaces at right angles. The sides and ends shall be machined square to the working surface. Plates of 10 inches through 12 inches ("X" dimensions) shall have at least 2 ribs, 16 inches through 24 inches ("X" dimensions) shall have at least 3 ribs, and 36-inch plates shall have at least 4 ribs. Plates over 75 pounds in weight (10 inches "X" dimensions and over) shall be provided with lifting holes in ribbing or equal means for mechanical lifting. They shall conform to table II and shall be similar to figure 1.

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TABLE II. *Type I, classes 1 and 2, right-angle plates or knees, plain and slotted*

Dimensions		
X ± 1/64 inch	Y ± 1/64 inch	Z ± 1/64 inch
Inches	*Inches*	*Inches*
2-1/2	3	4
4	4	6
5	5	5
5	4-1/2	5
6	8	8
6	6	8
10	10	10
10	8	12
12	12	12
16	18	20
24	24	24
36	36	36

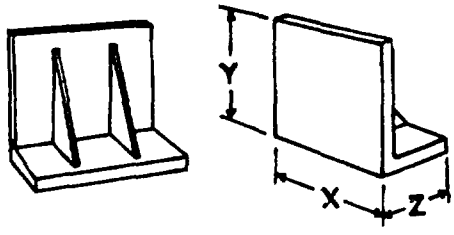


FIGURE 1. Type I, class 1, plain right-angle plates or knees.

3.10.2 *Class 2, slotted.* Class 2 angle plates or knees shall have two members with the outside working surfaces at right angles. Horizontal or vertical slots shall be provided in one or both members for clamping of the work. Slot sizes are optional with the manufacturer. However, the selection of slot sizes shall be in accordance with accepted manufacturing practice covering this product. The sides and ends shall be machined square with the working surfaces. Class 2 angle plates and knees shall conform to table II and shall be similar to figure 2.

3.10.3 *Class 3, measuring, plain.* Class 3 angle plates or knees shall have two members with outside working surfaces at right angles. Both working surfaces shall be finished in like manner and in accordance with the grade specified (see 6.2). Unless otherwise specified in the contract or order, the sides and ends shall be ground for finish only. A rib of sufficient thickness shall be located between the ends and shall be cored out in such a manner as to assure good hand gripping without sacrificing the stabilizing characteristics of the rib. The short member "Y" shall be provided with a suitable hole so that the measuring plate may be bolted to the table of a machine. Class 3 angle plates or knees shall conform to table III and shall be similar to figure 3.

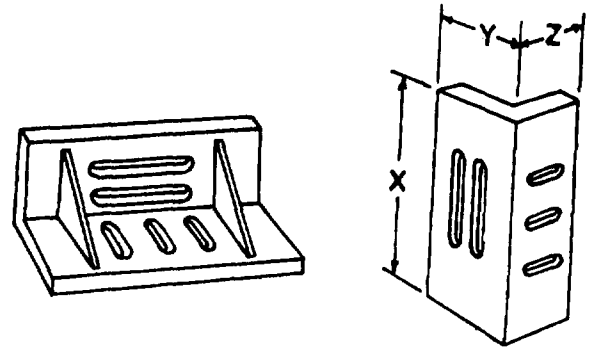


FIGURE 2. Type I, class 2, right-angle plates or knees, slotted.

TABLE III. *Type I, class 3, measuring, plain, right-angle plates or knees*

Dimensions		
X ± 1/16 inch	Y ± 1/16 inch	Z ± 5/16 inch
Inches	*Inches*	*Inches*
2-1/2	6	12-1/4
4	8	21-1/4

3.11 **Type II, inside right-angle plates.** The inside right-angle plate shall be as inverted T-shaped casting formed by a base and a perpendicular of approximately the same thickness. The one side of the perpendicular, the top side of the base adjacent and the bottom of the base shall be the working surfaces. All three working surfaces shall be finished in like manner and in accordance with the grade specified (see 6.2). The unfinished side of the perpendicular shall be supported by ribbing sufficient to assure stability and accuracy. The sides opposite and ends opposite shall be machined parallel to each other and square to their respective working surfaces. Type II angle plates or

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knees shall conform to table IV and shall be similar to figure 4.

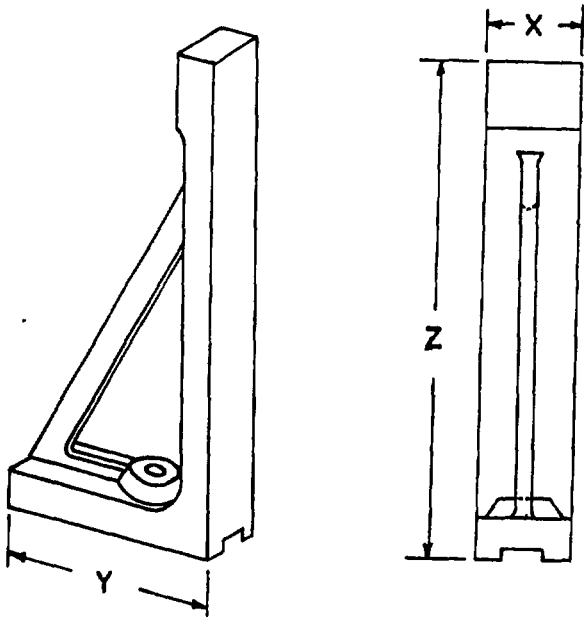


FIGURE 3. Type I, class 3, measuring, plain, right-angle plates or knees.

TABLE IV. *Type II, inside right-angle plates*

Dimensions		
X ± 1/64 inch	Y ± 1/64 inch	Z ± 1/64 inch
Inches	*Inches*	*Inches*
4	4	6
6	6	8
8	8	8
10	10	10
12	12	12
18	18	18

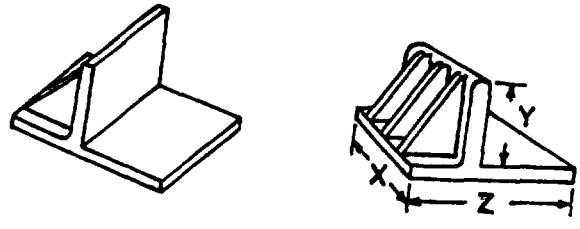


FIGURE 4. Type II, inside right-angle plates

3.12 Type III, universal right-angle plates. Universal right-angle plates shall have two working surfaces at right angles. All adjacent sides shall be square within the same limits that govern the squareness of the two working surfaces and all opposite sides shall be parallel within 0.0002 inch per foot. A rib of sufficient thickness shall be located between the ends and shall be cored out in such a manner as to assure good hand gripping without sacrificing the stabilizing characteristics of the rib. Type III angle plates shall conform to table V and shall be similar to figure 5.

TABLE V. *Type III, universal right-angle plates*

Dimensions		
X ± 1/64 inch	Y ± 1/64 inch	Z ± 1/64 inch
Inches	*Inches*	*Inches*
3	4	6
3-3/4	4	5
4	6	8
6	6	6
8	8	8
8	10	18
12	12	24
12	13	20

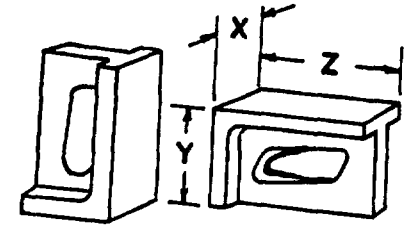


FIGURE 5. Type III, universal right-angle plates

3.13 Type IV machinists adjustable angle plates. Machinists adjustable angle plates shall consist essentially of a tilting table mounted on a turntable type base. Base shall be provided with heavy lugs with base slots for mounting on machine tools. The bottom of the base shall be machined and finished true flat, parallel and squared to the vertical axis. Top of the base shall be graduated in one degree increments for the full 360° and shall be marked at least every 10° for at least 90° each side of zero ("0"). Grad-

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uations and markings shall be permanently and legibly engraved or etched on a chrome or bright finished face. Noncumulative error between graduations shall not exceed 2-1/2 minutes. The tilting table shall consist of two plates joined together by a precision hinge and provided with an adjustable and positive locking mechanism. The bottom plate, which shall be attached to the base at the vertical axis, shall swivel a full 360° and shall be marked with a permanent and distinctive line for accurate positioning with graduations on the base. Top plate shall be a one piece casting with at least three "T" slots milled from the solid. "T" slots shall have at least 5/8-inch bolt capacity. Unless otherwise specified (see 6.2), working surface and bearing area of the top plate shall conform to grade C and table I. Top plate shall be vertically adjustable from zero horizontal to at least 90° vertical. Set adjustment shall be by means of an integral protractor graduated in one degree increments and marked at least every 10°. Noncumulative error between graduations shall not exceed 2-1/2 minutes. Type IV machinists adjustable angle plates shall be similar to figure 6 and shall conform to plate dimensions of table VI.

TABLE VI. *Type IV machinist's adjustable angle plate*

Dimensions	
X	Y
Inches	*Inches*
8	10
8	12
12	12
12	18

3.14 Type V, toolmaker's adjustable knee. The toolmaker's adjustable knee shall be constructed of a plate and a right-angle iron with two ribs. The right-angle iron and the plate are joined by a precision hinge and a lock mechanism. The plate shall swing to any position from zero (horizontal) to 90° (vertical) about the hinge axis and be capable of being securely locked in any position about the axis and shall be provided with T-slots for the mounting of work. The toolmaker's adjustable knee shall be equipped with a quadrant scale and a vernier scale. The scale shall be mounted in such a manner as to maintain parallactic error to a minimum.

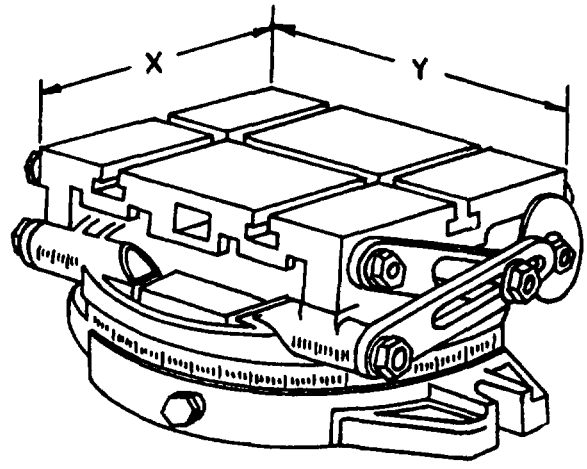


FIGURE 6. Type IV, machinists' adjustable-angle plate.

3.14.1 *Vernier scale.* The vernier scale shall be graduated in 5-minute increments from 0 minutes to 60 minutes both right and left of the zero index. The graduations shall be cut clearly and straight to a width of not less than 0.003 nor more than 0.006 inch and to a depth of not less than 0.001 inch nor more than 0.003 inch. The 0 minute and each succeeding 15-minute graduation shall be marked with the corresponding number and shall be no longer than the intermediate graduations. The intermediate graduations shall be readable to 5-minute increments. The vernier scale shall be accurate within plus or minus 1 minute between any two graduations.

3.14.2 *Quadrant scale.* The quadrant scale shall have 90 graduations. The zero and each 10° graduation shall be marked with the corresponding numeral. The 5° graduations shall be no longer than the intermediate graduations. A tolerance of plus or minus 2 minutes noncumulative between any two graduations shall be allowed. Width and

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depth of graduation lines shall be as specified for the vernier scale (see 3.14.1).

3.14.3 Type V toolmaker's adjustable knees shall conform to table VII and shall be similar to figure 7.

TABLE VII. *Type V, toolmaker's adjustable knee*

Size	Table dimensions		Angle dimensions		
	X ± 1/64 Inch	Y ± 1/64 inch	A	B	C
	Inches	*Inches*	*Inches*	*Inches*	*Inches*
7 by 10	10	7	7	7	10

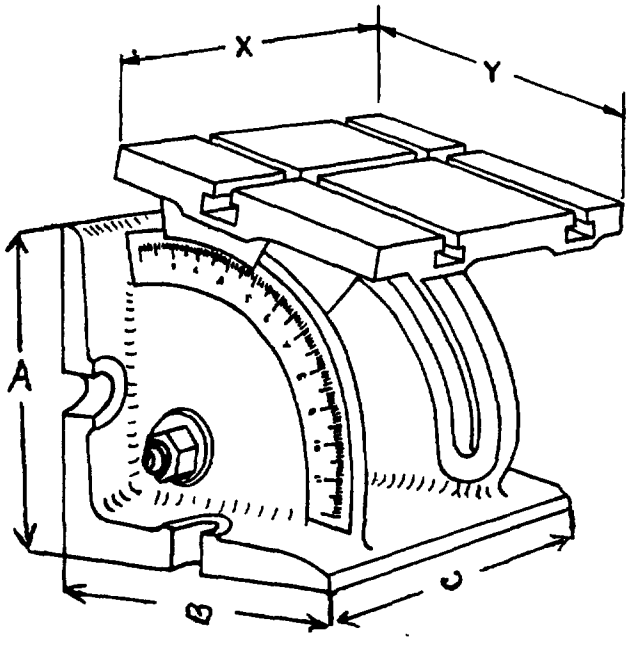


FIGURE 7. Type V, toolmaker's adjustable knee

3.15 Type VI, toolmaker's nonadjustable knees. Toolmaker's nonadjustable knees shall have two members with the outside working surfaces at right angles and with a finished pad on the back side of each member. The knee shall be designed with adequate thickness of section together with ribbing to insure rigidity and stability. The front side and the pad on the back side of each member shall be the working surfaces. The pads, sides opposite, and ends opposite shall be machined parallel to each other and square to their respective working surfaces, in accordance with the grade specified (see 3.6). Type VI toolmaker's knees shall conform to table VIII and shall be similar to figure 8.

TABLE VIII. *Type VI, toolmaker's nonadjustable knee*

Dimensions		
X ± 1/64 inch	Y ± 1/64 inch	Z ± 1/64 inch
Inches	*Inches*	*Inches*
2-1/2	8	4

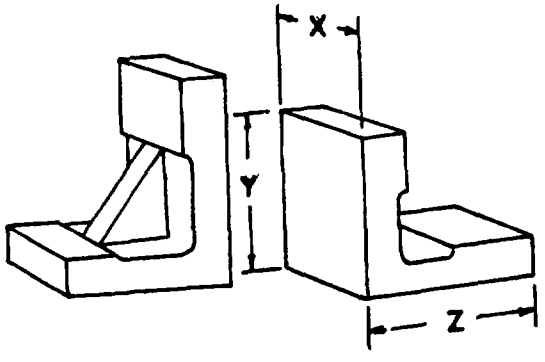


FIGURE 8. Type VI, toolmaker's nonadjustable knee.

3.16 Identification marking. Angle plates and knees shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.17 Workmanship. Workmanship of the angle plates and knees shall be of the quality prevailing among manufacturers normally producing precision type angle plates of the types specified herein.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for

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the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 *Lot.* All angle plates or knees of the same type, class, grade, and size presented at one time shall be considered a lot for purposes of inspection.

4.2.2 *Sampling for visual and dimensional examination.* A random sample of angle plates or knees shall be selected from each lot of material offered for examination of visual and dimensional characteristics in accordance with MIL-STD-105 at inspection level III for lots of 40 and under, inspection level II for lots of 41 to 300 and inspection level I for lots over 300. The acceptable quality level (AQL) shall be equal to 1.5 percent defective.

4.2.3 *Sampling for quality conformance tests.* A random sample of angle plates or knees shall be selected from each lot in accordance with MIL-STD-105 at inspection level S-4. The AQL shall be 2.5 percent defective. However, the sample size shall be the number associated with the letter in the table and the acceptance number shall be zero until the AQL permits one or greater.

4.3 Quality conformance inspection.

4.3.1 *Examination.* Each of the sample angle plates or knees selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in table IX. Any angle plate or knee in the sample containing one or more defects shall not be offered for delivery, and if the number of defective angle plates or knees in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

TABLE IX. *Classification of defects in accordance with MIL-STD-105*

Categories	Defects
Critical:	None defined.
Major:	
101	Type, class (when applicable), and grade of angle plates or knees not as specified.
102	Construction not free from any characteristics or defects that may render the angle plate or knee unsuitable for the purpose intended.
103	Castings not thoroughly cleaned of all other scale, molding sand, and other extraneous materials; evidence of flaws in castings repaired by plugging, peening, or welding except as noted.
104	All unmachined surfaces not painted as specified.
105	Grade A angle plates or knees nonconforming, working surfaces not fine precision scraped as specified; the spotting cavities depth, concentration and bearing area not within the specified minimum and maximum values; deviation of bearing area from the mean plane exceeds the specified maximum; variance of adjacent square areas from each other exceeds the specified maximum; variance of adjacent working surfaces of angle plates or knees from square exceeds the specified maximum.
106	Grade B angle plates or knees nonconforming, working surfaces not precision ground as specified; surface roughness (r.m.s.) not as specified for the specified sizes of angle plates or knees; deviation of the bearing area of each working surface from the mean plane exceeds the specified maximum; variance of adjacent square areas from each other exceeds the specified maximum; variance of adjacent working surfaces of angle irons and knees from square exceeds the specified maximum.
107	Grade C angle plates or knees nonconforming, working surfaces not smooth machined as specified; surface roughness (r.m.s.) not as specified; deviation of the bearing

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TABLE IX. *Classification of defects in accordance with MIL-STD-105 (cont'd)*

Categories	Defects
Critical:	
Major:	
107 (cont'd)	area of each working surface from the mean plane exceeds the specified maximum; variance of adjacent square areas from each other exceeds the specified maximum; variance of adjacent working surfaces of angle irons and knees from square exceeds the specified maximum.
108	Ribbing not incorporated into the design of angle plates and knees as specified.
109	Threaded parts (when applicable) nonconforming, form and class of fit not as specified; threads missing, stripped, crossed, torn or damaged.
110	Type I, class 1 plain angle plates or knees nonconforming, configuration not specified; outside working surfaces not at right angles; sides and ends not machined square to the working surfaces; dimensions not within the specified minimum and maximum tolerance.
111	Type I, class 2 slotted angle plates or knees nonconforming, configuration not as specified; horizontal or vertical slots not provided as specified; working surfaces not at right angles; sides and ends not machined square to the working surfaces; dimensions not within the specified minimum and maximum tolerance.
112	Type I, class 3, measuring, plain angle plates or knees nonconforming, configuration not as specified; outside working surfaces of the members not at right angle; working surfaces not finished as specified; sides and ends not ground for finish only except when otherwise specified; rib nonconforming or not provided as specified; bolt hole not provided in the short leg as specified; dimensions not within the specified minimum and maximum tolerances.
118	Type II, inside right angle plates nonconforming, configuration not as specified; working surfaces not as specified, not finished as specified; the unfinished side of the perpendicular not supported by ribbing as specified; the sides opposite and the ends opposite not machined parallel to each other and square to their respective working surfaces; dimensions not within the specified minimum and maximum tolerance.
114	Type III universal right angle plates nonconforming, configuration not as specified; working surfaces nonconforming, not at right angles; all adjacent sides not square within the specified limits; all opposite sides not parallel within the specified limits; rib not provided and located as specified; dimensions not within the specified minimum and maximum tolerance.
115	Type IV machinist's adjustable angle plates nonconforming, not composed of the specified parts; the adjustable angle plate not capable of being rotated and locked to any position about the vertical axis; adjustable angle plates not provided with T-slots as specified; adjustable angle plates not composed of two plates with a precision hinge and locking mechanism as specified; plate not fastened to the base as specified; failure of the other plate to be capable of adjustment as specified about the hinged axis and securely locking in any position; adjustable angle plate not equipped with protractor as specified; protractor marking not graduated as specified, tolerance of graduations not within the specified values, graduation lines not permanently marked or legible as specified; dimensions of adjustable angle plates not within the specified minimum and maximum tolerance.
116	Type V toolmaker's adjustable knee nonconforming, not composed of the specified parts; the right angle iron and the plate not joined together by a precision hinge and a lock mechanism; failure of the plate to be capable of swinging to and locking in any position about the hinge axis as specified; toolmaker's ad-

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TABLE IX. *Classification of defects in accordance with MIL-STD-105 (cont'd)*

Categories	Defects
Critical:	
Major:	
116 (cont'd)	justable knee not provided with T-slots and equipped with a guadrant scale and a vernier scale as specified, scale not mounted so as to maintain parallactic error to a minimum. Vernier scale not graduated as specified, graduations not cut clearly and straight, width and depth of graduations not within the specified minimum and maximum values; graduations not marked as specified, intermediate graduations not readable as specified; accuracy of vernier scale graduations not within the specified minimum and maximum values. Quadrant scale graduations not as specified, graduations not marked as specified; tolerance of graduations not within the specified minimum and maximum values. Dimensions of toolmaker's adjustable knees not within the specified minimum and maximum tolerance.
117	Type VI toolmaker's nonadjustable knees nonconforming, configuration not as specified; outside working surfaces not at right angles, pad not finished on back side of each member; ribbing not provided to insure rigidity and stability; pads, sides opposite and ends opposite not machined parallel to each other and square to their respective working surfaces; dimensions of toolmaker's nonadjustable knees not within the specified minimum and maximum tolerances.
118	Marking of manufacturer's name or trademark nonconforming, missing, or not permanent as specified.
Minor:	
201	Cleaning, preservation, packaging, packing, and marking of angle plates or knees nonconforming or not as specified.

4.3.2 *Tests.* Each of the sample angle plates or knees selected in accordance with 4.2.3 shall be tested in accordance with 4.4.1 and 4.4.2 to verify compliance with this specification. Any sample angle plate or knee which does not conform to these characteristics shall not be offered for delivery, and if the number of nonconforming angle plates or knees in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.4 Test procedures.

4.4.1 *Accuracy of working surfaces.* The accuracy of the working surfaces shall be determined by checking each working surface for flatness and by checking each adjacent working surface for squareness in accordance with the tolerances specified for the applicable grade (see 3.6). Methods used in determining conformance with this specification shall be an approved method standard with industry and acceptable to the procuring agency.

4.4.2 *Bearing areas.* The percentage of bearing area shall be determined by very lightly coating the working surface with Prussian blue or other suitable material, then rubbing over the surface with another plate of known accuracy to bring out the high spots. A plate of glass or other suitable flat transparent material that has previously been ruled into small predetermined squares shall then be laid on the working surface and the percent of bearing determined as well as the number of spots per square inch.

4.4.3 *Hardness test.* One sample angle plate from each lot offered shall be tested in conformance with method 242 of Fed. Test Method Std. No. 151, using a 3000 kilogram load and a 10 milliliter diameter ball. Readings to be taken on sections back of or opposite to the working surfaces. Preparation of the test surfaces shall be made in accordance with method 242 of Fed. Test Method Std. No. 151. Failure of the plate to meet the minimum requirements of 180 Brinell hardness number shall be cause for rejection of the lot represented.

4.4.4 *Reports.* When required (see 6.2), the contractor shall furnish reports on tests

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in duplicate showing the quantitative results for all tests and analysis of material used and signed by the director of the laboratory in which the tests were conducted or by his authorized assistant.

4.5 **Inspection of preparation for delivery requirements.** An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table X. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

TABLE X. *Classification of preparation for delivery defects*

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Cleaning and drying.* The plates shall be cleaned by process C-5 and dried by procedure D-1 in accordance with MIL-P-116.

5.1.1.2 *Preservative application.* All unpainted and unprotected surfaces of the plates subject to corrosion shall be coated with type P-8 preservative in accordance with MIL-P-116.

5.1.1.3 *Unit packaging.* All surfaces coated with preservative shall be covered with material conforming to MIL-P-3420, type II, class 1. A sheet of fiberboard meeting the requirements of PPP-F-320, covering the entire precision surfaces shall be utilized. The fiberboard shall be secured with tape conforming to PPP-T-76.

5.1.1.3.1 *Plates weighing 65 pounds or less.* Each plate weighing 65 pounds or less shall be packaged in a close-fitting box conforming to PPP-B-636, V3c, or PPP-B-640, grade A, class 2. Blocking and bracing shall be provided. The box shall be closed in accordance with the appendix to the box specification.

5.1.2 *Level B.* The plates shall be packaged as specified in 5.1.1 except the fiberboard box shall be the domestic type.

5.1.3 *Level C.* The plates shall be packaged in accordance with the suppliers commercial practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.*

5.2.1.1 *Plates weighing 65 pounds or less.* Unit packages of plates of like description weighing 65 pounds or less shall be packed in a close-fitting box conforming to PPP-B-601, type III or PPP-B-621, class 2. The box shall be closed and strapped in accordance with the appendix to the box specification. The gross weight shall not exceed 200 pounds.

5.2.1.2 *Plates weighing in excess of 65 pounds.* Plates weighing in excess of 65 pounds shall be individually packed in a close-fitting box conforming to PPP-B-601, type III or PPP-B-621, class 2. Blocking and bracing shall be provided. The box shall be

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closed and strapped in accordance with the appendix to the box specification. When the gross weight exceeds 250 pounds, skids shall be provided in accordance with the box specification. A crate conforming to MIL-C-104 shall be provided for plates whose net weight exceeds 500 pounds.

5.2.2 *Level B.* The plates shall be packed in accordance with 5.2.1 except the boxes shall be the domestic type.

5.2.3 *Level C.* The plates shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to marking required by the contract or order, the shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military activities.* In addition to markings required by the contract or order, the shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** Types I, II, and III angle plates or knees are intended to be used by craftsmen for ordinary machine shop operations. Types IV, V, and VI angle plates or knees are intended to be used principally by toolmakers and precision type work.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, and grade of plate or knee (see 1.2.1, 1.2.2 and 3.6).
(*c*) Size (see 1.2.3 and tables II through VIII).
(*d*) Specify if a particular color is desired (see 3.3.1).
(*e*) Special test reports, when required (see 4.4.4).
(*f*) Levels of preservation, packaging, and packing required (see 5.1 and 5.2).

6.3 **Transportation description.** Transportation descriptions and minimum weight applicable to this commodity are:

Rail:

Tools, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

Motor:

Tools, not otherwise indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—WC
Navy—SH
Air Force—67

Reviewer:

Navy—SH
Air Force—67

User:

Army—WC, MO, GL
Navy—WP, YD

CIVIL INTEREST:

National Bureau of Standards
Department of the Interior
Veterans Administration
General Services Administration

Preparing Activity:

Navy—SH

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